

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459526.7103 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1706 +/- 0.0088
Transit depth (Rp/Rs)^2	2.91 +/- 0.3 [%]
Orbital Inclination (inc)	89.79 +/- 1.52
Airmass coefficient 1 (a1)	1.061 +/- 0.012
Airmass coefficient 2 (a2)	-0.05 +/- 0.017
Scatter in the residuals of the lightcurve fit is	1.13 %
Best Comparison Star	#2 - [401, 223]
Optimal Aperture	4.21
Optimal Annulus	6.44
Transit Duration (day)	0.1324 +/- 0.0015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1706 ± 0.0088 vs 0.1489 (deviation 2.47σ)
Duration fit vs published	0.1324 d vs 0.1304 d (deviation 1.32σ)
Mid-time O–C	-8.8 min
Depth SNR	19.4
Residual scatter	1.13 %

Light curve

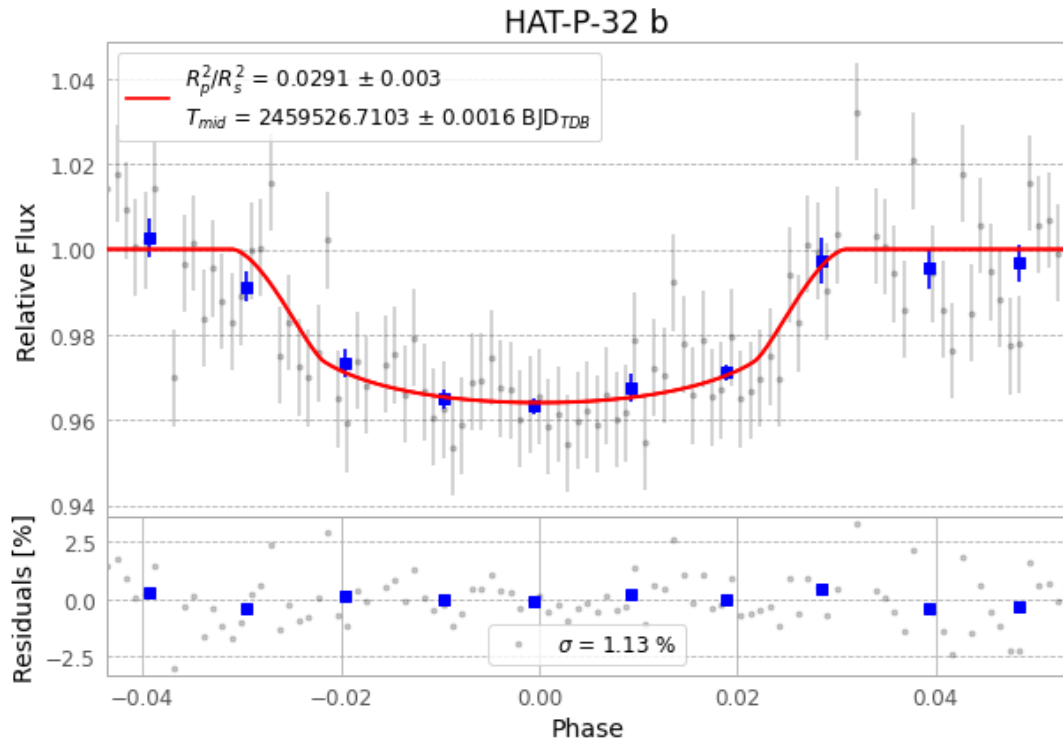


Figure 1 — FinalLightCurve_HAT-P-32 b_2021-11-07.png (SHA-256 849e9248a5a37fd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [370, 231], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 85b34f23dc07af5d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

101 FITS frames (66 MB), HATP-32211108023912.FITS → HATP-32211108073912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-211108.log (93b7de08305a...), FinalLightCurve_HAT-P-32 b_2021-11-07.png (849e9248a5a3...), FinalLightCurve_HAT-P-32 b_2021-11-07.csv (7ebd3604e9f4...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 18:22Z.